

Product Data Cleaning and Preparation Using Excel

Data Cleaning and Insights

Project Objective:

To perform data cleaning and preparation on a retail product sales dataset using Microsoft Excel. The goal is to handle missing values, correct inconsistent text formatting, remove duplicate records, split the Product ID into Manufacturing Date and Country Code, merge Brand Name and Product Name into Product Brand, apply currency and date formatting, and use conditional formatting to highlight patterns in the dataset.

Dataset Description:

The dataset is a retail product sales dataset containing 34 records across multiple categories including Electronics, Fashion, Kitchen, Outdoor, and Accessories. Products are sold across 12 countries — US, UK, Canada, India, Australia, Germany, Italy, France, Spain, China, Russia, and Brazil. Each record includes the product name, brand, selling price, quantity sold, and category. After cleaning the dataset was reduced to 32 rows by removing 2 duplicate records.

Column Description

Column Name	Description
Product ID	Unique identifier combining manufacturing date and country code (e.g. 28-JAN-US)
Product Name	Name of the product (e.g. Laptop, Sneakers, Blender)
Brand Name	Brand of the product (e.g. Dell, Nike, Ninja)
Price (\$)	Selling price of the product in US dollars — formatted as \$#,##0.00 currency
Quantity	Number of units sold
Category	Product category — Electronics, Fashion, Kitchen, Outdoor, Accessories
Manufacturing Date	Date extracted from Product ID in DD-MM-YYYY format — added in Column G using TEXT formula
Country Code	Country code extracted from Product ID using RIGHT formula — added in Column H
Product Brand	Brand Name and Product Name merged into one column — added in Column I using concatenation formula

Data Cleaning and Transformation:

The dataset did not contain any missing values in the Price or Category columns. The Product Name column had one inconsistency — T-shirt in Row 7 was corrected to T-Shirt using Find and Replace (Ctrl+H). The Category column was checked for typos and all five values were correctly spelled. Two exact duplicate rows were removed using Data then Remove Duplicates — Row 24 (duplicate of Row 16) and Row 33 (duplicate of Row 11) — reducing the dataset from 34 to 32 rows. The Product ID was split into Manufacturing Date (Column G) using

the TEXT DATEVALUE formula and Country Code (Column H) using RIGHT formula. A new Product Brand column (Column I) was created by merging Brand Name and Product Name. The Price column was formatted as \$#,##0.00 currency and conditional formatting was applied — data bars on Price and a yellow highlight for Electronics in Category.

Visualization and Insights Arrived with Screenshot of Visualization Report/Dashboard:

Descriptive Analysis:

The cleaned dataset contains 32 products after removing 2 duplicate records. The Price column is formatted as currency with values from \$30.00 to \$1,000.00. Three new derived columns were added — Manufacturing Date in DD-MM-YYYY format, Country Code, and Product Brand. Products span 5 categories and 12 countries.

Diagnostic Analysis:

One text inconsistency was found and corrected in the Product Name column. Two duplicate rows were removed. The Electronics category is the most frequent and is highlighted in yellow in the Category column. Data bars on the Price column show the price distribution across all products.

Predictive Analysis:

Data bars show that Laptops (\$1,000.00), Cameras (\$700.00), and Smartphones (\$900.00) carry the highest prices. These all belong to Electronics so maintaining good stock levels of Electronics products will keep revenue consistent.

Prescriptive Analysis:

The Manufacturing Date and Country Code columns enable analysis by region and time period. Consistent formatting and duplicate removal ensures accuracy. The Product Brand column simplifies product identification. Future data entry should follow the same formatting standards.

Reference

Fig 1 & Fig 2 — Task 1: Missing Values — Price Column | Category Column

Q1. Check for missing values in the Price column. How would you handle products with missing price information?

Answer: No missing values found in the Price column. If present, missing prices would be filled with the median price of the same category to maintain data consistency.

Q2. If there are products with missing categories, propose a strategy to impute or deal with these missing values effectively.

Answer: No missing Category values found. Strategy: infer from Product Name (e.g. Laptop → Electronics). If not inferable, assign Uncategorized as placeholder.

A	B	C	D	E	F
Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category
28-JAN-US	Laptop	Dell	\$1,000.00	30	Electronics
15-FEB-US	Sneakers	Nike	\$80.00	15	Fashion
03-MAR-US	Coffee Maker	Keurig	\$130.00	40	Kitchen
11-APR-US	Smartphone	Samsung	\$900.00	25	Electronics
22-MAY-US	Backpack	North Face	\$70.00	20	Outdoor
07-JUN-UK	Headphones	Sony	\$200.00	45	Electronics
19-JUL-UK	T-Shirt	Adidas	\$30.00	5	Fashion
23-AUG-UK	Blender	Ninja	\$90.00	35	Kitchen
05-SEP-UK	Tablet	Apple	\$500.00	50	Electronics
14-OCT-UK	Hiking Boots	Timberland	\$130.00	10	Outdoor
17-JUN-IN	Laptop	HP	\$950.00	25	Electronics
25-NOV-AU	Sneakers	Adidas	\$90.00	40	Fashion
08-DEC-DE	Coffee Maker	Nespresso	\$120.00	35	Kitchen
18-FEB-CA	Smartwatch	Fitbit	\$150.00	15	Electronics
16-APR-ES	Headphones	Bose	\$250.00	20	Electronics
21-AUG-CA	Laptop Bag	Samsonite	\$50.00	35	Accessories
20-AUG-CN	Smartwatch	Huawei	\$160.00	15	Electronics
27-JAN-IT	Laptop	Asus	\$980.00	10	Electronics
01-MAR-UK	Sunglasses	Oakley	\$150.00	15	Fashion
14-AUG-US	Camping Tent	Coleman	\$200.00	10	Outdoor
14-MAY-RU	Camera	Nikon	\$700.00	50	Electronics
09-JAN-CA	Microwave	Panasonic	\$80.00	20	Kitchen
19-JUL-BR	Fitness Tracker	Xiaomi	\$150.00	30	Electronics
29-SEP-CA	Smartphone	Google	\$800.00	45	Electronics
03-JUN-CA	Sunglasses	Ray-Ban	\$130.00	25	Fashion
11-JUL-CA	Blender	Vitamix	\$400.00	40	Kitchen
16-APR-ES	Headphones	Bose	\$230.00	20	Electronics
07-MAR-CA	Dress	Zara	\$60.00	30	Fashion
13-APR-CA	Toaster	Hamilton	\$40.00	10	Kitchen
24-MAY-CA	Fitness Tracker	Garmin	\$130.00	5	Electronics
02-DEC-CA	Jeans	Levi's	\$50.00	50	Fashion
09-JUL-FR	Watch	Casio	\$100.00	20	Accessories

Fig 1.1 – Price column showing no missing values (all 32 rows contain data)

A	B	C	D	E	F
Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category
28-JAN-US	Laptop	Dell	\$1,000.00	30	Electronics
15-FEB-US	Sneakers	Nike	\$80.00	15	Fashion
03-MAR-US	Coffee Maker	Keurig	\$130.00	40	Kitchen
11-APR-US	Smartphone	Samsung	\$900.00	25	Electronics
22-MAY-US	Backpack	North Face	\$70.00	20	Outdoor
07-JUN-UK	Headphones	Sony	\$200.00	45	Electronics
19-JUL-UK	T-Shirt	Adidas	\$30.00	5	Fashion
23-AUG-UK	Blender	Ninja	\$90.00	35	Kitchen
05-SEP-UK	Tablet	Apple	\$500.00	50	Electronics
14-OCT-UK	Hiking Boots	Timberland	\$130.00	10	Outdoor
17-JUN-IN	Laptop	HP	\$950.00	25	Electronics
25-NOV-AU	Sneakers	Adidas	\$90.00	40	Fashion
08-DEC-DE	Coffee Maker	Nespresso	\$120.00	35	Kitchen
18-FEB-CA	Smartwatch	Fitbit	\$150.00	15	Electronics
16-APR-ES	Headphones	Bose	\$250.00	20	Electronics
21-AUG-CA	Laptop Bag	Samsonite	\$50.00	35	Accessories
20-AUG-CN	Smartwatch	Huawei	\$160.00	15	Electronics
27-JAN-IT	Laptop	Asus	\$980.00	10	Electronics
01-MAR-UK	Sunglasses	Oakley	\$150.00	15	Fashion
14-AUG-US	Camping Tent	Coleman	\$200.00	10	Outdoor
14-MAY-RU	Camera	Nikon	\$700.00	50	Electronics
09-JAN-CA	Microwave	Panasonic	\$80.00	20	Kitchen
19-JUL-BR	Fitness Tracker	Xiaomi	\$150.00	30	Electronics
29-SEP-CA	Smartphone	Google	\$800.00	45	Electronics
03-JUN-CA	Sunglasses	Ray-Ban	\$130.00	25	Fashion
11-JUL-CA	Blender	Vitamix	\$400.00	40	Kitchen
16-APR-ES	Headphones	Bose	\$230.00	20	Electronics
07-MAR-CA	Dress	Zara	\$60.00	30	Fashion
13-APR-CA	Toaster	Hamilton	\$40.00	10	Kitchen
24-MAY-CA	Fitness Tracker	Garmin	\$130.00	5	Electronics
02-DEC-CA	Jeans	Levi's	\$50.00	50	Fashion
09-JUL-FR	Watch	Casio	\$100.00	20	Accessories

Fig 1.2 – Category column showing no missing values

Fig 3 & Fig 4 — Task 2: Inconsistent Data — Product Name Column | Category Column

Q3. Identify any inconsistent text formats present in the Product Name column.

Answer: T-shirt (Row 7, Product ID: 19-JUL-UK) used lowercase s while all other product names use Title Case. Corrected to T-Shirt using Find & Replace.

Q4. Identify any typos present in the Category column.

Answer: No typos found. All 5 category values — Electronics, Fashion, Kitchen, Outdoor, Accessories — are correctly spelled and consistent.

Q5. Use the Find and Replace function to standardize the text formats in the Product Name column and fix any typos or misspellings in the Category column.

Answer: Find & Replace (Ctrl+H): Find: T-shirt | Replace with: T-Shirt | Applied to Column B only. No replacements needed in Category column as all values were already correct.

A	B	C
Product ID	Product Name	Brand Name
28-JAN-US	Laptop	Dell
15-FEB-US	Sneakers	Nike
03-MAR-US	Coffee Maker	Keurig
11-APR-US	Smartphone	Samsung
22-MAY-US	Backpack	North Face
07-JUN-UK	Headphones	Sony
19-JUL-UK	T-Shirt	Adidas
23-AUG-UK	Blender	Ninja
05-SEP-UK	Tablet	Apple
14-OCT-UK	Hiking Boots	Timberland
17-JUN-IN	Laptop	HP
25-NOV-AU	Sneakers	Adidas
08-DEC-DE	Coffee Maker	Nespresso
18-FEB-CA	Smartwatch	Fitbit
16-APR-ES	Headphones	Bose
21-AUG-CA	Laptop Bag	Samsonite
20-AUG-CN	Smartwatch	Huawei
27-JAN-IT	Laptop	Asus
01-MAR-UK	Sunglasses	Oakley
14-AUG-US	Camping Tent	Coleman
14-MAY-RU	Camera	Nikon
09-JAN-CA	Microwave	Panasonic
19-JUL-BR	Fitness Tracker	Xiaomi
29-SEP-CA	Smartphone	Google
03-JUN-CA	Sunglasses	Ray-Ban
11-JUL-CA	Blender	Vitamix
16-APR-ES	Headphones	Bose
07-MAR-CA	Dress	Zara
13-APR-CA	Toaster	Hamilton
24-MAY-CA	Fitness Tracker	Garmin
02-DEC-CA	Jeans	Levi's
09-JUL-FR	Watch	Casio

Fig 2.1 – Product Name column — T-Shirt corrected to Title Case (Row 7)

A	B	C	D	E	F
Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category
28-JAN-US	Laptop	Dell	\$1,000.00	30	Electronics
15-FEB-US	Sneakers	Nike	\$80.00	15	Fashion
03-MAR-US	Coffee Maker	Keurig	\$130.00	40	Kitchen
11-APR-US	Smartphone	Samsung	\$900.00	25	Electronics
22-MAY-US	Backpack	North Face	\$70.00	20	Outdoor
07-JUN-UK	Headphones	Sony	\$200.00	45	Electronics
19-JUL-UK	T-Shirt	Adidas	\$30.00	5	Fashion
23-AUG-UK	Blender	Ninja	\$90.00	35	Kitchen
05-SEP-UK	Tablet	Apple	\$500.00	50	Electronics
14-OCT-UK	Hiking Boots	Timberland	\$130.00	10	Outdoor
17-JUN-IN	Laptop	HP	\$950.00	25	Electronics
25-NOV-AU	Sneakers	Adidas	\$90.00	40	Fashion
08-DEC-DE	Coffee Maker	Nespresso	\$120.00	35	Kitchen
18-FEB-CA	Smartwatch	Fitbit	\$150.00	15	Electronics
16-APR-ES	Headphones	Bose	\$250.00	20	Electronics
21-AUG-CA	Laptop Bag	Samsonite	\$50.00	35	Accessories
20-AUG-CN	Smartwatch	Huawei	\$160.00	15	Electronics
27-JAN-IT	Laptop	Asus	\$980.00	10	Electronics
01-MAR-UK	Sunglasses	Oakley	\$150.00	15	Fashion
14-AUG-US	Camping Tent	Coleman	\$200.00	10	Outdoor
14-MAY-RU	Camera	Nikon	\$700.00	50	Electronics
09-JAN-CA	Microwave	Panasonic	\$80.00	20	Kitchen
19-JUL-BR	Fitness Tracker	Xiaomi	\$150.00	30	Electronics
29-SEP-CA	Smartphone	Google	\$800.00	45	Electronics
03-JUN-CA	Sunglasses	Ray-Ban	\$130.00	25	Fashion
11-JUL-CA	Blender	Vitamix	\$400.00	40	Kitchen
16-APR-ES	Headphones	Bose	\$230.00	20	Electronics
07-MAR-CA	Dress	Zara	\$60.00	30	Fashion
13-APR-CA	Toaster	Hamilton	\$40.00	10	Kitchen
24-MAY-CA	Fitness Tracker	Garmin	\$130.00	5	Electronics
02-DEC-CA	Jeans	Levi's	\$50.00	50	Fashion
09-JUL-FR	Watch	Casio	\$100.00	20	Accessories

Fig 2.2 – Category column showing all values are correctly spelled

Fig 5 — Task 3: Removing Duplicate Records

Q6. Identify any duplicate rows within the dataset based on the entirety of each row, and remove them if any.

- Row 24 was a duplicate of Row 16 (21-AUG-CA | Laptop Bag | Samsonite | \$50 | 35 | Accessories)
- Row 33 was a duplicate of Row 11 (17-JUN-IN | Laptop | HP | \$950 | 25 | Electronics)

Answer: 2 exact duplicate rows removed using Data → Remove Duplicates (all 6 columns). Dataset: 34 rows → 32 rows.

	A	B	C	D	E	F
	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category
1	28-JAN-US	Laptop	Dell	\$1,000.00	30	Electronics
2	15-FEB-US	Sneakers	Nike	\$80.00	15	Fashion
3	03-MAR-US	Coffee Maker	Keurig	\$130.00	40	Kitchen
4	11-APR-US	Smartphone	Samsung	\$900.00	25	Electronics
5	22-MAY-US	Backpack	North Face	\$70.00	20	Outdoor
6	07-JUN-UK	Headphones	Sony	\$200.00	45	Electronics
7	19-JUL-UK	T-Shirt	Adidas	\$30.00	5	Fashion
8	23-AUG-UK	Blender	Ninja	\$90.00	35	Kitchen
9	05-SEP-UK	Tablet	Apple	\$500.00	50	Electronics
10	14-OCT-UK	Hiking Boots	Timberland	\$130.00	10	Outdoor
11	17-JUN-IN	Laptop	HP	\$950.00	25	Electronics
12	25-NOV-AU	Sneakers	Adidas	\$90.00	40	Fashion
13	08-DEC-DE	Coffee Maker	Nespresso	\$120.00	35	Kitchen
14	18-FEB-CA	Smartwatch	Fitbit	\$150.00	15	Electronics
15	16-APR-ES	Headphones	Bose	\$250.00	20	Electronics
16	21-AUG-CA	Laptop Bag	Samsonite	\$50.00	35	Accessories
17	20-AUG-CN	Smartwatch	Huawei	\$160.00	15	Electronics
18	27-JAN-IT	Laptop	Asus	\$980.00	10	Electronics
19	01-MAR-UK	Sunglasses	Oakley	\$150.00	15	Fashion
20	14-AUG-US	Camping Tent	Coleman	\$200.00	10	Outdoor
21	14-MAY-RU	Camera	Nikon	\$700.00	50	Electronics
22	09-JAN-CA	Microwave	Panasonic	\$80.00	20	Kitchen
23	19-JUL-BR	Fitness Tracker	Xiaomi	\$150.00	30	Electronics
24	29-SEP-CA	Smartphone	Google	\$800.00	45	Electronics
25	03-JUN-CA	Sunglasses	Ray-Ban	\$130.00	25	Fashion
26	11-JUL-CA	Blender	Vitamix	\$400.00	40	Kitchen
27	16-APR-ES	Headphones	Bose	\$230.00	20	Electronics
28	07-MAR-CA	Dress	Zara	\$60.00	30	Fashion
29	13-APR-CA	Toaster	Hamilton	\$40.00	10	Kitchen
30	24-MAY-CA	Fitness Tracker	Garmin	\$130.00	5	Electronics
31	02-DEC-CA	Jeans	Levi's	\$50.00	50	Fashion
32	09-JUL-FR	Watch	Casio	\$100.00	20	Accessories

Fig 3.1 – Cleaned dataset after duplicate removal (32 rows remaining)

Fig 6 & Fig 7 — Task 4: Splitting & Merging — Manufacturing Date + Country Code | Product Brand

Q7. Split the Product ID column into two separate columns for Manufacturing Date and Country Code. Remove unnecessary characters, if any.

- Manufacturing Date (Col G): =TEXT(DATEVALUE(LEFT(A2,6)&"-2024"),"DD-MM-YYYY")
- Country Code (Col H): =RIGHT(A2,2)

Answer: Product ID (format: DD-MMM-CC) split into Manufacturing Date (e.g. 28-01-2024) and Country Code (e.g. US, UK, IN) using Excel formulas.

Q8. Merge the Brand Name and Product Name columns into one column named Product Brand.

Answer: Product Brand (Col I): =C2&" "&B2 — combines Brand Name and Product Name (e.g. Dell Laptop, Nike Sneakers)

Manufacturing Date
28-01-2024
15-02-2024
03-03-2024
11-04-2024
22-05-2024
07-06-2024
19-07-2024
23-08-2024
05-09-2024
14-10-2024
17-06-2024
25-11-2024
08-12-2024
18-02-2024
16-04-2024
21-08-2024
20-08-2024
27-01-2024
01-03-2024
14-08-2024
14-05-2024
09-01-2024
19-07-2024
29-09-2024
03-06-2024
11-07-2024
16-04-2024
07-03-2024
13-04-2024
24-05-2024
02-12-2024
09-07-2024

Fig 4.1 – Manufacturing Date (DD-MM-YYYY) and Country Code columns extracted from Product ID

Manufacturing Date	Country Code	Product Brand
28-01-2024	US	Dell Laptop
15-02-2024	US	Nike Sneakers
03-03-2024	US	Keurig Coffee Maker
11-04-2024	US	Samsung Smartphone
22-05-2024	US	North Face Backpack
07-06-2024	UK	Sony Headphones
19-07-2024	UK	Adidas T-Shirt
23-08-2024	UK	Ninja Blender
05-09-2024	UK	Apple Tablet
14-10-2024	UK	Timberland Hiking Boots
17-06-2024	IN	HP Laptop
25-11-2024	AU	Adidas Sneakers
08-12-2024	DE	Nespresso Coffee Maker
18-02-2024	CA	Fitbit Smartwatch
16-04-2024	ES	Bose Headphones
21-08-2024	CA	Samsonite Laptop Bag
20-08-2024	CN	Huawei Smartwatch
27-01-2024	IT	Asus Laptop
01-03-2024	UK	Oakley Sunglasses
14-08-2024	US	Coleman Camping Tent
14-05-2024	RU	Nikon Camera
09-01-2024	CA	Panasonic Microwave
19-07-2024	BR	Xiaomi Fitness Tracker
29-09-2024	CA	Google Smartphone
03-06-2024	CA	Ray-Ban Sunglasses
11-07-2024	CA	Vitamix Blender
16-04-2024	ES	Bose Headphones
07-03-2024	CA	Zara Dress
13-04-2024	CA	Hamilton Toaster
24-05-2024	CA	Garmin Fitness Tracker
02-12-2024	CA	Levi's Jeans
09-07-2024	FR	Casio Watch

Fig 4.2 – Product Brand column showing merged Brand Name and Product Name

Fig 8 & Fig 9 — Task 5: Number Formatting — Currency Format | Date Format

Q9. Format the data type of the Price column to a currency format.

Answer: Format Cells (Ctrl+1 → Number → Currency): \$#,##0.00 applied to Column D. Values display as \$1,000.00, \$80.00 etc.

Q10. Format the Manufacturing Date column to display dates in the DD-MM-YYYY format.

Answer: Manufacturing Date (Col G) uses formula =TEXT(DATEVALUE(LEFT(A2,6)&"-2024"),"DD-MM-YYYY") producing DD-MM-YYYY output (e.g. 28-JAN-US → 28-01-2024)

A	B	C	D
Product ID	Product Name	Brand Name	Price (\$)
28-JAN-US	Laptop	Dell	\$1,000.00
15-FEB-US	Sneakers	Nike	\$80.00
03-MAR-US	Coffee Maker	Keurig	\$130.00
11-APR-US	Smartphone	Samsung	\$900.00
22-MAY-US	Backpack	North Face	\$70.00
07-JUN-UK	Headphones	Sony	\$200.00
19-JUL-UK	T-Shirt	Adidas	\$30.00
23-AUG-UK	Blender	Ninja	\$90.00
05-SEP-UK	Tablet	Apple	\$500.00
14-OCT-UK	Hiking Boots	Timberland	\$130.00
17-JUN-IN	Laptop	HP	\$950.00
25-NOV-AU	Sneakers	Adidas	\$90.00
08-DEC-DE	Coffee Maker	Nespresso	\$120.00
18-FEB-CA	Smartwatch	Fitbit	\$150.00
16-APR-ES	Headphones	Bose	\$250.00
21-AUG-CA	Laptop Bag	Samsonite	\$50.00
20-AUG-CN	Smartwatch	Huawei	\$160.00
27-JAN-IT	Laptop	Asus	\$980.00
01-MAR-UK	Sunglasses	Oakley	\$150.00
14-AUG-US	Camping Tent	Coleman	\$200.00
14-MAY-RU	Camera	Nikon	\$700.00
09-JAN-CA	Microwave	Panasonic	\$80.00
19-JUL-BR	Fitness Tracker	Xiaomi	\$150.00
29-SEP-CA	Smartphone	Google	\$800.00
03-JUN-CA	Sunglasses	Ray-Ban	\$130.00
11-JUL-CA	Blender	Vitamix	\$400.00
16-APR-ES	Headphones	Bose	\$230.00
07-MAR-CA	Dress	Zara	\$60.00
13-APR-CA	Toaster	Hamilton	\$40.00
24-MAY-CA	Fitness Tracker	Garmin	\$130.00
02-DEC-CA	Jeans	Levi's	\$50.00
09-JUL-FR	Watch	Casio	\$100.00

Fig 5.1 – Price column formatted as \$#,##0.00 currency

Manufacturing Date
28-01-2024
15-02-2024
03-03-2024
11-04-2024
22-05-2024
07-06-2024
19-07-2024
23-08-2024
05-09-2024
14-10-2024
17-06-2024
25-11-2024
08-12-2024
18-02-2024
16-04-2024
21-08-2024
20-08-2024
27-01-2024
01-03-2024
14-08-2024
14-05-2024
09-01-2024
19-07-2024
29-09-2024
03-06-2024
11-07-2024
16-04-2024
07-03-2024
13-04-2024
24-05-2024
02-12-2024
09-07-2024

Fig 5.2 – Manufacturing Date column displaying dates in DD-MM-YYYY format

Fig 10 & Fig 11 — Task 6: Conditional Formatting — Data Bars on Price | Electronics Highlight

Q11. Apply data bar or color scales conditional formatting in the Price column.

Answer: Home → Conditional Formatting → Data Bars → Blue Data Bar applied to D2:D33. Longer bars = higher price, shorter bars = lower price.

Q12. Create a custom rule for conditional formatting in the Category column to highlight cells where the category is Electronics.

Answer: Home → Conditional Formatting → New Rule → Cell Value = Electronics | Format: Yellow fill + Bold font | Applied to F2:F33

A	B	C	D
Product ID	Product Name	Brand Name	Price (\$)
28-JAN-US	Laptop	Dell	\$1,000.00
15-FEB-US	Sneakers	Nike	\$80.00
03-MAR-US	Coffee Maker	Keurig	\$130.00
11-APR-US	Smartphone	Samsung	\$900.00
22-MAY-US	Backpack	North Face	\$70.00
07-JUN-UK	Headphones	Sony	\$200.00
19-JUL-UK	T-Shirt	Adidas	\$30.00
23-AUG-UK	Blender	Ninja	\$90.00
05-SEP-UK	Tablet	Apple	\$500.00
14-OCT-UK	Hiking Boots	Timberland	\$130.00
17-JUN-IN	Laptop	HP	\$950.00
25-NOV-AU	Sneakers	Adidas	\$90.00
08-DEC-DE	Coffee Maker	Nespresso	\$120.00
18-FEB-CA	Smartwatch	Fitbit	\$150.00
16-APR-ES	Headphones	Bose	\$250.00
21-AUG-CA	Laptop Bag	Samsonite	\$50.00
20-AUG-CN	Smartwatch	Huawei	\$160.00
27-JAN-IT	Laptop	Asus	\$980.00
01-MAR-UK	Sunglasses	Oakley	\$150.00
14-AUG-US	Camping Tent	Coleman	\$200.00
14-MAY-RU	Camera	Nikon	\$700.00
09-JAN-CA	Microwave	Panasonic	\$80.00
19-JUL-BR	Fitness Tracker	Xiaomi	\$150.00
29-SEP-CA	Smartphone	Google	\$800.00
03-JUN-CA	Sunglasses	Ray-Ban	\$130.00
11-JUL-CA	Blender	Vitamix	\$400.00
16-APR-ES	Headphones	Bose	\$230.00
07-MAR-CA	Dress	Zara	\$60.00
13-APR-CA	Toaster	Hamilton	\$40.00
24-MAY-CA	Fitness Tracker	Garmin	\$130.00
02-DEC-CA	Jeans	Levi's	\$50.00
09-JUL-FR	Watch	Casio	\$100.00

Fig 6.1 – Price column with blue data bars showing relative price values

A	B	C	D	E	F
Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category
28-JAN-US	Laptop	Dell	\$1,000.00	30	Electronics
15-FEB-US	Sneakers	Nike	\$80.00	15	Fashion
03-MAR-US	Coffee Maker	Keurig	\$130.00	40	Kitchen
11-APR-US	Smartphone	Samsung	\$900.00	25	Electronics
22-MAY-US	Backpack	North Face	\$70.00	20	Outdoor
07-JUN-UK	Headphones	Sony	\$200.00	45	Electronics
19-JUL-UK	T-Shirt	Adidas	\$30.00	5	Fashion
23-AUG-UK	Blender	Ninja	\$90.00	35	Kitchen
05-SEP-UK	Tablet	Apple	\$500.00	50	Electronics
14-OCT-UK	Hiking Boots	Timberland	\$130.00	10	Outdoor
17-JUN-IN	Laptop	HP	\$950.00	25	Electronics
25-NOV-AU	Sneakers	Adidas	\$90.00	40	Fashion
08-DEC-DE	Coffee Maker	Nespresso	\$120.00	35	Kitchen
18-FEB-CA	Smartwatch	Fitbit	\$150.00	15	Electronics
16-APR-ES	Headphones	Bose	\$250.00	20	Electronics
21-AUG-CA	Laptop Bag	Samsonite	\$50.00	35	Accessories
20-AUG-CN	Smartwatch	Huawei	\$160.00	15	Electronics
27-JAN-IT	Laptop	Asus	\$980.00	10	Electronics
01-MAR-UK	Sunglasses	Oakley	\$150.00	15	Fashion
14-AUG-US	Camping Tent	Coleman	\$200.00	10	Outdoor
14-MAY-RU	Camera	Nikon	\$700.00	50	Electronics
09-JAN-CA	Microwave	Panasonic	\$80.00	20	Kitchen
19-JUL-BR	Fitness Tracker	Xiaomi	\$150.00	30	Electronics
29-SEP-CA	Smartphone	Google	\$800.00	45	Electronics
03-JUN-CA	Sunglasses	Ray-Ban	\$130.00	25	Fashion
11-JUL-CA	Blender	Vitamix	\$400.00	40	Kitchen
16-APR-ES	Headphones	Bose	\$230.00	20	Electronics
07-MAR-CA	Dress	Zara	\$60.00	30	Fashion
13-APR-CA	Toaster	Hamilton	\$40.00	10	Kitchen
24-MAY-CA	Fitness Tracker	Garmin	\$130.00	5	Electronics
02-DEC-CA	Jeans	Levi's	\$50.00	50	Fashion
09-JUL-FR	Watch	Casio	\$100.00	20	Accessories

Fig 6.2 – Category column with Electronics cells highlighted in yellow bold

Conclusion:

This assignment successfully demonstrated data cleaning and preparation on a retail product sales dataset using Microsoft Excel. All six tasks were completed — missing value handling, inconsistent data correction, duplicate removal, column splitting and merging, number formatting, and conditional formatting. The dataset was reduced from 34 to 32 rows by removing 2 duplicates. One text inconsistency was corrected and three new columns were added. The final cleaned dataset is well structured, consistently formatted, and ready for further analysis.